

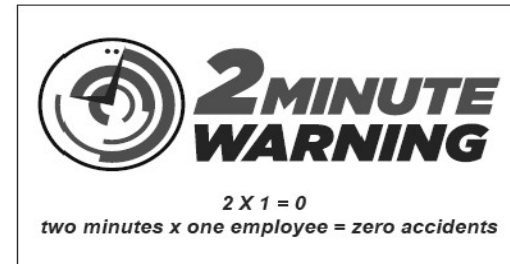
DESIGN GUIDELINES FOR PIPES, HOSES AND CLIPS
PROCEDURE


BEFORE CONTINUING, PLEASE ENSURE YOU HAVE READ AND UNDERSTOOD THE RELEVANT S.O.P.

This document has been compiled from information gathered from results of Technical Reports, previous published guidelines and practical observations.

The purpose of this document is to aid designers by providing guidance to;

1. Outline the applications where each type of clip is used
2. Detail recommended hose and clip combinations, installation torque values.
3. Clarify recommended interference between hoses, pipes and clips.
4. Detail installation guidelines
5. Stub pipe design, flared "fishtail" and parallel section pipes.
6. Accessory pipe design.
7. Pipe hose retention swage profile design.
8. Hose length calculation.

ATTENTION POINTS

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1. HOSE AND CLIP APPLICATIONS.

Hose type	Recommended clip type
EPDM braided rubber (Black with white braiding)	Norma worm drive clip, Norma T-clip, Norma "Torro" constant torque worm drive clip
Silicon nomex reinforced rubber (Red or blue)	Norma T-clip, Breeze Constant-torque or High-torque clip.
APT thick walled rubber (Extruded polymer, black)	Breeze Constant-torque clip

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2. RECOMMENDED HOSE AND CLIP COMBINATIONS AND TIGHTENING TORQUE.
2.1. Norma worm drive clip

Part No.	Dia. range (mm)	Pipe dia. (mm)	Installation torque (Nm)	Socket required (mm)	Hose type
36001	8 – 16	1/2" & 3/8" hose (Aeroquip fittings)	3 (+0.5Nm)	7	EPDM rubber
36005	12 – 20	12.7	3 (+0.5Nm)	7	EPDM rubber
36007	16 – 25		4.5 (+1.0Nm)	7	EPDM rubber
36009	25 – 40	32	4.5 (+1.0Nm)	7	EPDM rubber
36013	20 – 32	25mm hose. (Aeroquip fittings)	4.5 (+1.0Nm)	7	EPDM rubber
36014	32 – 50	38	4.5 (+1.0Nm)	7	EPDM rubber
36015	40 – 60	50.8	4.5 (+1.0Nm)	7	EPDM rubber
36019	50 – 70	57	4.5 (+1.0Nm)	7	EPDM rubber
36021	60 – 80	63	4.5 (+1.0Nm)	7	EPDM rubber
36023	70 – 90	76	4.5 (+1.0Nm)	7	EPDM rubber
36025	80 - 100	85 & 89	4.5 (+1.0Nm)	7	EPDM rubber
36027	90 – 110	Test tanks only	4.5 (+1.0Nm)	7	EPDM rubber
36029	100 – 120	Test tanks only	4.5 (+1.0Nm)	7	EPDM rubber
36031	120 - 140	Test tanks only	4.5 (+1.0Nm)	7	EPDM rubber

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2.2. Norma "Torro" constant torque worm drive clip

Part No.	Dia. Range (mm)	Pipe dia. (mm)	Installation torque (Nm)	Socket required (mm)	Hose type
3600001400	20 – 32	25	5 (+0.5)	7	EPDM rubber
3600001200	50 – 70	57	5 (+0.5)	7	EPDM rubber
3600001300	60 – 80	63	5 (+0.5)	7	EPDM rubber

2.3. Norma T-clip

Part No.	Dia. range (mm)	Pipe dia. (mm)	Installation torque (Nm)	Bolt size (mm)	Hose type
32200/04	43 - 47	38.1	4	M7 X 50	EPDM rubber
32200/04	43 – 47	38.1	4	M7 x 50	Silicone nomex
32200/06	51 - 55	44.4	4	M7 x 60	EPDM rubber
32200/07	59-63	50.8	4	M7 x 60	EPDM rubber
32200/08	63 – 68	57.1/60.3	4	M7 x 60	EPDM rubber
32200/09	68 – 73	63.5	20	M8 x 90	EPDM rubber
32200/10	73 - 79	69	20	M8 x 90	EPDM rubber
32200/11	79 – 85	76.2	20	M8 x 90	EPDM rubber
32200/13	97 - 104	88.9	20	M8 x 90	EPDM rubber
32200/13	97 – 104	88.9	20	M8 x 90	Silicone nomex
32200/16	121 – 130	114.3	20	M8 x 90	EPDM rubber
32200/16	121 – 130	114.3	20	M8 x 90	Silicone nomex
32200/18	130 – 140	127	20	M8 x 100	EPDM rubber
32200/18	130 – 140	127	20	M8 x 100	Silicone nomex
32200/21	162 - 174	152.4	30	M10 x 110	Silicone nomex
3600001900	91-97	85.0	12	M8 x 90	EPDM rubber

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2.4. Breeze Constant-torque and High-torque clip

Part No.	Vendor part No.	Dia. range (mm)	Pipe dia. (mm)	Installation torque (Nm)	Socket required (ins.)	Hose type
32229/01	CT175L	25.4 – 44.4	25.4	8	3/8	EPDM rubber
32229/02	CT200L	31.7 – 54.1	38.1	14	3/8	APT thick wall
32229/02	CT200L	31.7 – 54.1	38.1	14	3/8	EPDM rubber
32229/02	CT200L	31.7 – 54.1	38.1	14	3/8	Silicone nomex
32229/04	CT300L	57.1 – 79.5	57.1	14	3/8	EPDM rubber
32229/04	CT300L	57.1 – 79.5	57.1	14	3/8	APT thick wall
32229/05	CT350L	69.8 – 92.2	76.2	14	3/8	APT thick wall
32229/05	CT350L	69.8 – 92.2	76.2	14	3/8	Silicone nomex
32229/05	CT350L	69.8 – 92.2	76.2	14	3/8	EPDM rubber
32229/06	CT400L	82.5 – 104.9	88.9	14	3/8	APT thick wall
32229/07	CT450L	95.2 – 117.6	101.6	14	3/8	APT thick wall
32229/07	CT450L	95.2 – 117.6	101.6	14	3/8	Silicone nomex
32229/07	CT450L	95.2 – 117.6	101.6	14	3/8	EPDM rubber
32229/09	CT-550L	120.60-143	127	14	8mm	Silicone nomex
32229/10	CT600L	133.3 – 155.7	127	14	3/8	APT thick wall
3600000600	HD350	70 - 92	76.2	14	8 mm	Silicone nomex
3600000700	HD400	83 - 105	88.9	14	8mm	Silicone nomex
3600000800	HD450	95 - 118	101.6	14	8 mm	Silicone nomex
3600000900	HD550	121 - 143	114.3	14	8 mm	Silicone nomex

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